

BlueJammer-V2 (BW16 Controller) — Complete Hardware Guide

Firmware: BlueJammer-V2 — BW16 controller half · **Upstream:** [EmenstaNougat/BlueJammer-V2](#) (closed-source, pre-compiled bins) **Chip:** RTL8720DN (Ai-Thinker/B&T **BW16**, dual-band 2.4/5 GHz WiFi + BLE 5.0) · **Cyber Controller profile:** bluejammer-bw16 (slug bluejammer-bw16, rtl8720 backend, AmebaD ImageTool, pinned SHA-256 bundle, no suicide) **This guide:** purchase → build → **flash-and-study only** → integrate into Cyber Controller → lab handling → troubleshoot. ■ **LAB-ONLY / ILLEGAL TO OPERATE** — see §2 before doing anything.

1. Overview

BlueJammer-V2 is a **two-board RF-research platform** by Emensta: an **ESP32-WROOM-32U jamming engine** (with up to 4× nRF24L01+ modules and a 128×64 OLED) paired with an **Ai-Thinker BW16 (RTL8720DN) controller**. This guide covers **only the BW16 controller half** — the chip Cyber Controller actually flashes.

The BW16's job in this platform is **control, not RF emission**: it hosts a self-contained 5 GHz WiFi access point and a web UI, acts as the **UART master** to the ESP32 engine, and bridges the engine's telemetry (mode, nRF module status, serial log) into that web page. The four jamming *modes* (Bluetooth / BLE / WiFi / RC-Drone) are executed by the ESP32 + nRF24 side, not the BW16. The RTL8720DN is a dual-core part (KM4 Cortex-M33 @200 MHz + KM0) on Realtek's **AmebaD** platform, which is why it is flashed with Realtek's AmebaD image toolchain rather than esptool.

Cyber Controller's role is narrow and deliberate: it **flashes the pinned, SHA-256-verified AmebaD firmware bundle to the BW16 via the rt18720 backend, and nothing more**. There is **no serial command channel** on the controller firmware, so Cyber Controller **cannot and does not operate, key, or transmit** anything. The device's only operating surface is its own web UI + physical button — documented here, **not driven**.

2. Legal & Safety — READ THIS FIRST

■ **RF JAMMING IS ILLEGAL TO OPERATE. THIS IS A FLASH-AND-STUDY ARTIFACT ONLY.**

Operating an RF jammer is a **federal crime in the United States** under **47 U.S.C. § 333** ("No person shall willfully or maliciously interfere with or cause interference to any radio communications of any station licensed or authorized...") and is prohibited by **FCC rules (Parts 2/15)** — the FCC states jammers may not be operated, marketed, or sold, with penalties including **fines exceeding \$100,000, equipment seizure, and imprisonment**. Jamming is **similarly illegal in virtually every other jurisdiction worldwide** (e.g. UK Wireless Telegraphy Act, EU radio regulations, Canada, Australia). Even *possession* or *import* is restricted in some places.

What this guide authorizes: flashing the BW16 controller firmware and **studying the device on a bench**. Nothing more.

What this guide does NOT do — by design: - It gives **no transmit, keying, or jamming instructions**. The "modes" are listed only to document the control surface. - Cyber Controller has **no operate/transmit control** over this profile (danger: `illegal-tx`, no serial command channel). It flashes; it does not run the jammer.

If you choose to power the assembled platform at all, do so **only inside a fully enclosed, authorized RF-shielded environment** (a properly rated **RF anechoic/shielded chamber or Faraday enclosure** with no leakage) under an institution's spectrum/lab authorization. **Do not transmit over the air, ever**. Energising the nRF24 stage outside such an enclosure — even briefly, even "just to test" — is the illegal act. When in doubt, **keep the RF stage unpowered** and study the BW16 controller/web UI in isolation.

This guide is for education, security research, and lab study by people who will obey the law. You are solely responsible for compliance.

3. Hardware & Purchasing

You only need the **BW16 controller module** for the part Cyber Controller flashes. The rest (ESP32, nRF24s) belongs to the RF engine and is intentionally out of scope here.

Part	What to get	Why	Where to buy (search terms)
Controller (this guide)	Ai-Thinker / B&T BW16 (RTL8720DN) module	The chip CC flashes; dual-band 2.4/5 GHz WiFi + BLE 5.0, KM4+KM0 cores	Search " BW16 RTL8720DN module " on AliExpress/Amazon; also " BW16 Type-C RTL8720DN dev board ". Maker: Ai-Thinker (B&T)
USB-UART (if bare module)	CP2102 / CH340 USB-TTL adapter (3.3 V)	BW16 flashes over UART, not native USB	AliExpress/Amazon: "CP2102 USB to TTL 3.3V"
Data cable	Quality USB data cable (not charge-only)	Needed for the USB-Type-C BW16 variant / for the UART adapter	any reputable vendor

Notes & verify-before-you-buy: - The Cyber Controller profile records this board as **bw16-rtl8720dn**, flash size **2MB**. Some BW16 variants (e.g. **BW16 Type-C / BW16E**) ship with more flash — **verify: confirm your specific module's flash size and pinout against its datasheet**; the AmebaD bundle here targets the standard BW16. - Genuine **emensta / official source**: upstream warns "*If you purchased this device from anyone other than official sources at emensta.pages.dev — you have been scammed.*" **Verify the seller** before buying a pre-assembled BlueJammer-V2; this guide does not endorse any reseller. - **Avoid invented links**. Use the search strings above; vendor URLs and stock change constantly — confirm at purchase time (§9).

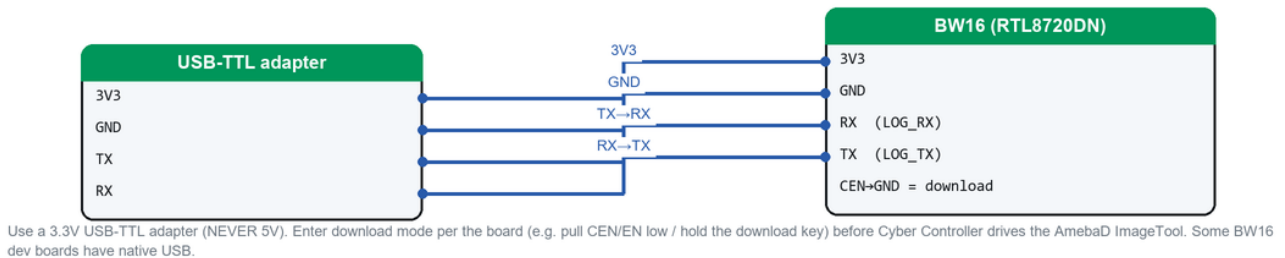
4. Building / Assembly (controller side)

The BW16 here is the **UART-master controller**. Assembly relevant to *flashing and studying the controller*:

- **Bare BW16 module:** wire your USB-UART adapter to the BW16's **LOG_UART** pins (cross **TX↔RX**), **3V3**, and **GND**. The **BW16 Type-C** variant has onboard USB and needs only a data cable.
- **Within the full BlueJammer-V2 platform** (documented for completeness — RF stage stays unpowered outside a shielded lab):
- BW16 ↔ ESP32 link is **UART @ 115200 baud, TX/RX crossed**: ESP32 TX (GPIO33 → PB2) to BW16 RX, ESP32 RX (GPIO26 ← PB1) to BW16 TX. A 2 s heartbeat / **LINKOK** handshake runs over this link.
- Power: *"Both boards share 3.3 V and GND. The BW16 can be powered directly from the ESP32's 3V3 rail."*
- The nRF24 + ESP32 jamming engine (quad nRF24L01+, 10 µF decoupling per module mandatory, OLED on I²C) is the **RF-emitting** half — **not** flashed by this profile and **not to be energised outside a shielded chamber** (§2).
- **Drivers (host):** install the USB-serial driver for your adapter/board — **CP210x** or **CH340** — or no COM port appears.

5. Flashing & First Run (via Cyber Controller)

Flashing connection — BW16 / RTL8720DN (UART)



How to connect the BW16/RTL8720DN over a 3.3V USB-TTL adapter for AmebaD flashing.

Cyber Controller flashes the BW16 controller using the `rtl8720 backend` (Realtek **AmebaD ImageTool / upload_image_tool**), the same flow as the standalone `rtl8720` profile.

One-time host prerequisite (AmebaD tool): the `rtl8720 backend` needs **Realtek's upload_image_tool** on the host (it is **not bundled** — closed/GPL constraints). Provide it via the `CYBERC_AMEBAD_TOOL` env var, or drop it in `tools/realtek/` (on **Windows** include `cygwin1.dll` alongside it).

1. Connect the BW16 by USB (Type-C variant) or via the USB-UART adapter; open Cyber Controller → **Flash** tab.
2. **Port:** select the BW16's COM/tty (click *Refresh* if missing → driver issue).
3. **Firmware Profile:** `BlueJammer-V2 – BW16 controller [LAB-ONLY / illegal to operate]` (id `bluejammer_bw16`, core `bluejammer-bw16`).
4. **Enter download mode on the BW16** when prompted: **hold BURN** → **press & release RST** → **release BURN**. (The AmebaD loader can also assert it via DTR/RTS on supported wiring.)
5. Click **Flash**. Cyber Controller, via the `rtl8720 backend`: - Fetches the **pinned v0.2 release bundle** — `km0_boot_all.bin`, `km4_boot_all.bin`, `km0_km4_image2.bin` — plus the shared `imgtool_flashloader_amebad.bin` SRAM loader (reused from the `rtl8720` bundle). - **Verifies every file's SHA-256** before writing (see §6 for the pinned digests). A mismatch aborts the flash. - Uploads the SRAM flashloader, then writes the AmebaD images (app offset `0x0`, merged-single-bin image model). Monitor/CLI baud is **115200**; the AmebaD loader negotiates **~1.5 Mbaud** for the transfer itself.
6. **First run:** after reset the BW16 brings up its self-hosted AP `BlueJ-V2_by_@emensta` on a 5 GHz channel (default **149**). Confirming the controller is alive (it advertises its AP) is the extent of "first run" here — **do not select a jamming mode / do not key the RF engine** (§2).

6. Integrate into Cyber Controller

- **Profile:** bluejammer-bw16 — backend: rtl8720, protocol: bluejammer, danger: illegal-tx, supports_suicide: false, resolver: pinned_release (tag: v0.2), app_offset: 0x0, image_model: merged-single-bin.
- **Firmware source:** EmenstaNougat/BlueJammer-V2, pinned release **v0.2** (.../releases/download/v0.2); the SRAM loader comes from the rtl8720 raw bundle. Firmware is **closed-source**: bins are **pinned and SHA-256-verified, never vendored** into the repo.
- **Pinned SHA-256 digests** (what CC enforces):
 - km0_boot_all.bin → 453c880307fc389009aa8a39c63da3d715b207ad797206c896cc691426daaa64
 - km4_boot_all.bin → 05fbf808d43113eaf7c11b75091c986b817135f0c52eb9fa94bb9fe9f34b062c
 - km0_km4_image2.bin → 615c382d48b89e1faceeba0ac586538894a21d4eeb4dc0c64027fcb271a84ef9
 - imgtool_flashloader_amebad.bin → 9307121385cb390dfd2da64da2c6c515f17b5a9556b3d04021487c9b9f220b55
- (Note: km0_km4_image2.bin here is the BlueJammer-V2 v0.2 image — a different digest from the standalone rtl8720 Vampire-Deauther image; the boot files and SRAM loader are shared.)
- **Control surface — documented, NOT driven:** the device is operated only from its **self-hosted web UI** (join AP BlueJ-V2_by_@emensta / pass NoConn1337 → http://192.168.1.1) plus the physical button. The web UI exposes mode buttons (**Bluetooth, BLE, WiFi, RC/Drone, Idle**), nRF module status, a live serial monitor/command log, OLED brightness (0–100 %), and AP timeout/shutdown settings. **There is no serial command channel**, so Cyber Controller **cannot key the transmitter** — and, per the illegal-to-operate framing, **does not**. Any future in-app control panel would have to POST to the web UI's mode endpoints (which must be captured on real hardware first) and would be **gated behind the illegal-tx confirmation**.
- **No Dead Man's Switch / suicide:** supports_suicide: false for this profile.
- **Backup first:** use **Backup** in the Flash tab to dump the BW16's current flash before overwriting.

7. Usage (lab study only — NO transmit)

This section is intentionally **not** an operating manual. There are **no jamming instructions** here.

1. **Flash & verify (the supported action):** flash the controller per §5; confirm CC reports success and the SHA-256 checks passed.
2. **Confirm the controller booted:** with the **RF engine unpowered**, the BW16 should advertise its AP `BlueJ-V2_by_@emensta`. Seeing the AP / reaching `http://192.168.1.1` confirms the controller firmware is healthy. **Leave the mode set to Idle.**
3. **Study the web UI as an artifact:** inspect the page structure, telemetry fields, and link/handshake (`LINKOK`) behaviour between BW16 and ESP32 — **without selecting a jamming mode and without energising the nRF24 stage.**
4. **Do not transmit.** Selecting Bluetooth/BLE/WiFi/RC-Drone and powering the RF stage outside a properly authorized, fully enclosed RF-shielded chamber is the **illegal act** (§2). If you have no such facility, **keep the RF stage unpowered** — study the controller in isolation.
5. **Re-flash / restore:** to return the BW16 to other firmware, re-flash via the appropriate profile (e.g. the standalone `rt18720` profile) using the same AmebaD flow.

8. Troubleshooting

- **No COM port:** install the **CP210x/CH340** driver for your adapter/BW16; try another **data** cable; on Linux add your user to `dialout` and replug.
- **upload_image_tool not found / flash won't start:** set **CYBERC_AMEBAD_TOOL** or place the tool in `tools/realtek/` (Windows: include `cygwin1.dll`). This is the most common AmebaD failure.
- **Flash fails to connect / loader won't upload:** re-enter download mode — **hold BURN, press & release RST, release BURN** — then retry; close any serial monitor holding the port; verify TX/RX are **crossed** on a bare module.
- **SHA-256 mismatch / download aborts:** CC refuses to flash unverified bins — a corrupted download or an upstream asset change is the usual cause; clear the cache and retry, and **verify the release tag is still v0.2**.
- **AP BlueJ-V2_by_@emensta not visible after flash:** the AP is on a **5 GHz channel (default 149)** — use a 5 GHz-capable client; confirm the controller actually booted (re-flash if download mode was missed mid-write).
- **Wrong-flash / boot issues:** confirm the module is a genuine **BW16 RTL8720DN** and **verify its flash size** (profile assumes 2MB); a mismatched variant can fail to boot.
- **Bricked controller:** re-run the AmebaD flash from download mode; the 3-file bundle + SRAM loader is the full image set, so a clean re-flash normally recovers it.

9. Sources

- Upstream firmware: <https://github.com/EmenstaNougat/BlueJammer-V2> (README, Firmware-Flasher, v0.2 release) — closed-source, pre-compiled bins. Official device source per upstream: emensta.pages.dev. Related upstream repos: [EmenstaNougat/ESP32-BlueJammer](#), [EmenstaNougat/BW16-BlueJammer](#).
- Cyber Controller profile: `src/config/profiles/bluejammer_bw16.json` (backend, pinned v0.2 assets + SHA-256, control surface, danger flag) and `src/config/profiles/rtl8720.json` (shared AmebaD backend behaviour, baud, download mode).
- BW16 / RTL8720DN module: Realtek AmebaD platform; Ameba Arduino docs and BW16 troubleshooting guides (forum.amebaiot.com), [pushtoop/BW16_RTL8720](#), Hackster.io BW16 getting-started/troubleshooting. **Verify your module's exact flash size and pinout against its datasheet.**
- Legal: **47 U.S.C. § 333** and FCC jammer-enforcement guidance (FCC Parts 2/15); equivalent prohibitions worldwide. Confirm current law in your jurisdiction before any RF work.
- Verify board availability, prices, the upstream release tag/digests, and current law at the time of use — links, stock, and statutes change.